

Towards HealthDCAT-AP compliant provenance description of clinical trial data flows according to ICH E6

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Abstract

Many researchers consider the provenance of data sets to be very important when it comes to trust and reusability of medical data sets. While the topic is being intensively studied in academia, information on data origin is often lacking in practice. One reason for this is the lack of coordinated, easily applicable recommendations for guidelines to produce provenance statements that can be interpreted comparatively across data sets. The HealthDCAT-AP vocabulary was developed to describe data sets in national catalogs and enables simple as well as complex provenance statements using the W3C PROV model. The objective of this work is to utilize HealthDCAT-AP and PROV-O to trace lineage from a clinical trial data collection according to the ICH E6 guideline. For this, the data flows of an exemplary clinical trial were implemented. Since the adoption of Revision 3 of the ICH E6 guideline, data flow diagrams should be included as part of sponsors' data management plans. The results show that data flows can be generated and visualized straightforwardly with PROV. However, there is a lack of coordinated vocabularies for the types of actors, activities, and entities that occur in such data flows to make data flow diagrams easy to read across studies and to enable queries, for example, on the characteristics of data collection.

Keywords

provenance, EHDS, PROV-O, DCAT, HealthDCAT-AP

1. Introduction and

The provenance of a data set is generally understood to refer to statements about the origin of the data and data transformation processes. Provenance is often explicitly required, e.g., by the FAIR Data Principles [1]. The reason for this is that identical data sets in medical research must be interpreted differently. In contrast to clinical documentation, billing data deliberately contains gaps and simplifications in diagnoses. Estimated or self-reported observations of body weight are subject to understandable deviations when compared to measured weight.

The European Health Data Space (EHDS) [2] is an initiative based on a European Union regulation with a binding legal framework for the establishment and networking of large national data catalogs, which enable interested researchers to search for and request data sets. All holders of medical data are required to enter all their data sets into the catalogs and to update the entries. The requested personal data sets are made available for analysis by the designated stakeholders in so-called Secure Processing Environments in a complicated and costly process and cannot be accessed directly. It is obvious that the success of the EHDS will depend largely on the quality of the data set descriptions.

HealthDCAT-AP [3], a vocabulary based on DCAT-AP and W3C DCAT, was developed to describe the characteristics and contents of medical data sets. A vocabulary is currently being designed in the QUANTUM project [4] to label data quality. Both vocabularies contain explicit requirements for labeling provenance using the W3C PROV data model [5]. However, they do not specify the type and scope, which can lead to provenance descriptions that vary greatly in detail and are difficult to compare.

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The ICH guideline for good clinical practice (GCP) is a long-established framework for conducting clinical trials with human subjects [6]. In July 2025, the third revision became binding, imposing more extensive requirements on the documentation of data handling. One of these aspects concerns the representation of all data collection and processing in a “data flow diagram”. Goal of this work is to create machine-readable provenance statements in PROV and a graphical representation as a directed graph for a given data flow diagram.

2. Methods

Based on a graphical template [7], the data flows of a complex clinical trial were modeled and visualized in PROV-O. Various subtypes were classified for activities, entities, and agents.

3. Results

The purpose of a data flow diagram is to provide domain experts with an overview of where data is collected utilizing various types of information systems and how the data is processed. Critical paths in activities should also be identifiable, e.g. the chain for rapid reporting of serious adverse events to the regulatory authorities in the case of new medications in therapies.

In the first step, all artefacts of the reference template [7] were assigned PROV-O top-level categories such as agent, activity and entity. In some cases, the assignment was not without controversy. For reasons of consistent presentation, information systems such as electronic data collection systems or laboratory information systems were modelled as agents (n=8) alongside organisations such as the sponsor or the study centre. Entities (n=12) are all data (sets) and data stores such as the recruitment list, while activities (n=27) are the data flows and named processes such as the generation of analysis data sets in accordance with the SDTM standard. The graphical visualisation was automatically generated and laid out using the PROV Toolbox [8].

4. Discussion

The results achieved are only partially convincing. It was possible to map all the information objects from the template and generate machine-readable provenance information using the constructs specified in HealthDCAT-AP and QUANTUM from PROV-O. The idea of labelling certain types of data flows, such as manual versus electronic data entry or particularly critical paths, requires extensions to the vocabulary. A visualisation with a “flow” view can be generated automatically, but it is inferior to manually laid out graphics, which can also hide certain details in favour of clarity.

The question arises as to how helpful such complex information ultimately is for machine-actionable provenance if it is designed for each data set according to its own criteria, its own modelling conventions and varying levels of detail. Given that the EHDS aims to manage many distributed data catalogues, each containing tens of thousands of metadata records from a wide variety of health domains, it will not be realistically feasible for the average data user to familiarise themselves with the respective provenance model for every data record that comes into question during a data search.

There is therefore an urgent demand for best practice guidelines on which information systems, stakeholders and data processing methods are most relevant for researchers who wish to reuse data sets. Furthermore, there is a need for agreement on overarching vocabularies, e.g. for types of information systems, for typical roles in the clinical trial process and for typical activities such as screening, recruitment, data collection and analysis. This would enable cross-dataset queries. This goal is part of the current efforts of experts who have come together in the MIRAPIE initiative [9].

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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